

Composition: Perform Conditional Object Composition for All N Possible Decompositions



A. Jointly generate “person”, “cat” and “apple”.



B. Construct mask M to **preserve** **objects** and define editable regions.

Stage 1: Jointly generate $N-1$ of objects.

For diffusion timesteps $t = T, \dots, 0$



A. Apply layout guidance to place dog in editable part :

$$z_t \leftarrow z_t - \eta \nabla_{z_t} E(A, M, j'), \quad E(A, M, j') = \left(1 - \frac{\sum_u A_{uj'} M_u}{\sum_u A_{uj'}}\right)^2$$

B. Aggregate with generated objects:

$$z_t = M \odot z_t + (1 - M) \odot z_t^{\mathcal{O}'}$$

Stage 2: compose the remaining object conditioned on $N-1$ objects.